

Software Requirements Specification (SRS)

Project: Field to Marketplace – Farm Produce Marketplace with Harvest Pre-Booking System

1. Introduction

A web application that connects farmers directly with buyers. Farmers can list available crops, publish upcoming harvests, and manage orders, while buyers can purchase fresh produce or reserve future harvests through a secure platform.

2. Problem Statement

Farmers often depend on middlemen, receive unfair prices, and struggle to estimate demand before harvesting. Buyers also find it difficult to purchase fresh produce directly from trusted farmers.

3. Objectives

- Direct farmer-to-buyer marketplace
- Harvest pre-booking
- Reduce crop wastage
- Transparent order tracking
- Weather assistance

4. Users

Farmer • Buyer • Administrator

5. Functional Requirements

Registration/Login, role-based access, crop management, harvest pre-booking, browse/search products, place orders, order tracking, weather information, email notifications, admin management.

6. Unique Features

- Harvest Pre-Booking for upcoming crops.
- Weather API for farming decisions.
- Email notifications for orders and booking updates.
- Direct farmer-to-buyer marketplace without middlemen.

7. Non-Functional Requirements

Responsive UI, secure authentication, REST APIs, scalable architecture, reliable PostgreSQL storage, maintainable code.

8. Technology Stack

React.js • Spring Boot (Gradle) • Spring Data JPA • PostgreSQL • Spring Security • IntelliJ IDEA • Weather API • Email API

9. Database Tables

Users, Farmers, Buyers, Admin, Crops, HarvestPreBookings, Orders, Notifications.

10. Future Enhancements

Online payments, product ratings, multilingual support, AI crop suggestions, mobile application.